

Popularising Summary

Louis Vandenbruwaene

Master's thesis: an extremal problem on connectivity in networks (Erdős Problem 915).

Many of the networks we rely on are useful precisely because they offer backup routes. If one road is closed, traffic takes another. If one cable fails, a message still reaches its destination along a different chain of links. Mathematicians study this kind of structure with graphs, where dots stand for objects and lines stand for the connections between them. This thesis asks a simple-sounding question about such networks: how many connections can a network with a fixed number of points have, if no two of those points may be joined by too many separate, non-overlapping routes?

That question is a version of a problem posed decades ago by Béla Bollobás and Paul Erdős, listed today as Erdős Problem 915. In its original form it concerns ordinary networks and routes that may not share a connection, and for that case the answer has been known since a theorem of Mader. This thesis starts from that result and then changes the rules of the network one at a time: connections may be doubled up, directions may be added so that a link works only one way, or a single connection may join more than two points at once. With each change the pencil-and-paper proof becomes longer and more delicate, until checking it by hand stops being realistic.

At that point the thesis hands the work to a computer. One program describes every version of the network in the same way, measures exactly how many separate routes run between any two points, and proves the answer outright for small networks by checking every possibility. Where checking everything is hopeless it searches for good examples instead, imitating the way hot metal cools into an orderly shape, drifting at first and then locking into place. Reassuringly, when the search is pointed at the cases we already understand, it rediscovers the very networks the earlier proofs had singled out.

The interplay between proof and experiment then pays off in four places. For networks whose connections may each tie several points together at once, the thesis settles the two strictest limits on backup routes. Under the strictest one the answer is exact at every size. Under the next one the same formula is always an upper limit, and is reached exactly once the network has room to spare, which the thesis pins down precisely. For one-way networks it proves, with no assumption about what the best network looks like, that the largest possible number of links is a quarter of the square of the number of points, give or take an amount growing only in proportion to that number. That settles the shape of an answer whose leading term had been only a conjecture, and what remains unknown there is a single numerical factor. For one-way networks whose links may be doubled up, the thesis settles the exact answer completely. The idea is to find, inside any such network, a thin skeleton that alone remembers who can reach whom. Removing it costs every pair at least one backup route, so peeling off one skeleton for each unit of the limit accounts for every link and proves the count outright. And for one-way networks with many-point connections, the obvious two-layer design turns out to be the best there is for large networks.